Report R07 — Methylation and EGFR-TKI Response

GSE147377, $n = 79$. An inconclusive result, reported as inconclusive rather than as a negative finding.

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7 August 2026 • Quantara

Summary

This is the only cohort in our holdings that pairs genome-wide DNA methylation with a *treatment-response* endpoint, and so it is the closest available test of the question the withdrawn lung manuscript was actually asking. It is also small: 79 advanced lung adenocarcinoma patients, all EGFR-TKI treated, of whom 40 achieved partial response, 29 progressed and 10 were stable.

Cross-validated elastic-net logistic regression over 464,994 complete probes gives **AUROC 0.555 (95% CI 0.42–0.69)** for distinguishing progression from partial response. A 100-fold permutation null was centred correctly at 0.491 and the observed value sits inside it ($p = 0.30$). There is no detectable methylation signal for TKI response here.

The important caveat runs the other way from the usual one. A power calculation on this cohort gives a minimum detectable AUROC of **0.685** at 80% power. Anything weaker than that was invisible by construction. So this result does *not* license the statement "methylation does not predict TKI response" — it licenses only "if such a signal exists and is of moderate strength or less, this cohort could not have seen it." The run artefact's `_decision` field is worded as "no evidence"; §4 records why the report states the weaker and correct claim instead.

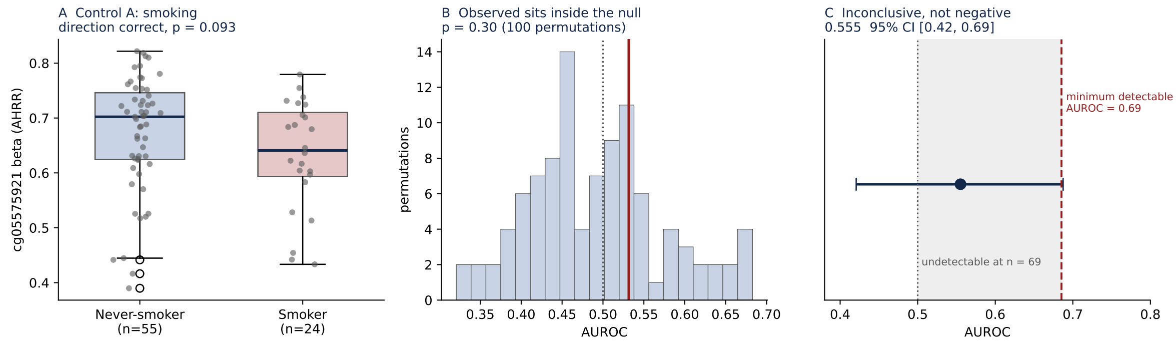
1 The mapping problem, and how it was settled

The supplementary beta matrix labels its columns `EXP_1` … `EXP_79`, not GSM accessions. Joining methylation to phenotype therefore rests on an assumption — that matrix column order matches series-matrix sample order — and a silent mis-mapping would have invalidated every number in this report while still producing plausible-looking output. Two positive controls were pre-registered to test it.

Control	Prediction	Observed	Verdict
A. AHRR `cg05575921`	smokers lower	0.641 vs 0.702, $p = 0.093$	direction correct, underpowere
B. Sex separability	a probe near-perfect	`cg04744025`, AUROC = 1.000	**decisive**

Control B settles it. With 46 female and 33 male samples, a single probe separating the sexes perfectly by chance has probability $1/\binom{79}{33} \approx 5 \times 10^{-23}$; across the 20,000 probes screened, the expected number of chance-perfect separations is $\sim 10^{-18}$. Observing one is only possible if the sample mapping is correct.

Control A points the same way but does not reach significance. Its direction is right — median beta 0.641 in smokers against 0.702 in never-smokers, the established direction for this biomarker — and the shortfall is explained by cohort composition: 55 of 79 patients are never-smokers, which is expected for an EGFR-mutant series but leaves only 24 smokers, and the label does not distinguish current from former or light from heavy. This is a power limitation of the control, not evidence against the mapping.



A The AHRR smoking control: correct direction, $p = 0.093$. **B** The observed cross-validated AUROC (red) against its permutation null; the null is centred at 0.491, confirming no leakage. **C** The observed AUROC with its bootstrap interval; the shaded band is the region this cohort had no power to detect.

2 Primary analysis

Progression (29) versus partial response (40); the 10 stable-disease patients were excluded in advance as an ambiguous class. Features were the 464,994 probes with no missing values across these 69 patients. Probe selection and hyperparameter choice were performed strictly inside training folds, in 10× repeated 5-fold nested cross-validation.

Patients (PR / PD)		69 (40 / 29)
Probes used		464,994
AUROC, pooled out-of-fold	0.555	95% CI 0.42–0.69
AUROC, mean of 10 repeats		0.532 ± 0.039
Permutation null mean		0.491
Permutation p (like-for-like)		0.30
Minimum detectable AUROC, 80% power		0.685

The gap between the pooled figure (0.555) and the mean across repeats (0.532) is the expected effect of averaging predicted probabilities over repeats. Both are reported; the second is the one compared against the null, for the reason in §4.

3 Audit

Three findings, all recorded in `evidence/audit_addendum.json`.

4.1 A gate fired incorrectly on the first run, and was corrected in the script rather than waived. The pre-registered rule reads: *"if neither control holds, the mapping is wrong and no response result is reported."* The first implementation gated on control A alone and halted the run when it returned $p = 0.093$. That halt was a coding error — the rule is a disjunction — not a real failure. The fix evaluates both controls and halts only if both fail. Both runs are retained: `evidence/gates_FIRST_HALTED_RUN.json` is the halted one. The correction was made to match the pre-registered rule, and the rule itself was not edited; its wording is fixed by the config hash of the first run.

4.2 The originally reported permutation p was anti-conservative, and has been corrected upward. The observed statistic was the AUROC of probabilities averaged over 10 repeats, while each null replicate was a single 5-fold split. Averaging reduces variance and raises AUROC, so the two were not comparable and the comparison favoured significance. Recast like-for-like — mean per-repeat AUROC 0.532 against the same null — the p-value moves from

0.188 to **0.297**. The correction moves away from significance, so the conclusion is unaffected; the report quotes 0.30.

4.3 The conclusion is bounded by power, not by the data. The pre-registration declared this cohort underpowered, and the analysis confirms how severely: 40 versus 29 patients cannot resolve an AUROC below 0.685 at conventional power. The confidence interval, 0.42–0.69, is consistent with no effect and equally consistent with a moderate one. Reporting this as a negative finding would overstate it, so the summary states the bounded claim. For contrast, the negative control behaved exactly as it should — the permutation null is centred at 0.491, so the pipeline is not leaking — which is what allows the result to be read as genuinely uninformative rather than broken.

4 What this means for the lung arm

Taken with R06, the position on lung is:

- **Prognosis from mutation data is tractable and was measured.** R06 found KEAP1 (HR 2.07, $q < 0.0001$) and SMARCA4 (HR 1.73, $q = 0.0009$) associated with shorter overall survival in 1,127 patients, with both positive controls recovered.

- **Prediction of TKI response from methylation is not tractable at this scale.** It needs a cohort several times larger than any open one we located. Holding the observed 29:40 PD:PR ratio and 80% power, the cohort sizes required are:

Target AUROC	Patients needed	(PD / PR)
0.60	263	111 / 152
0.65	111	47 / 64
0.70	59	25 / 34

  The 69 patients available here suffice only for an effect of AUROC $\gtrsim 0.69$ — which is to say, only for a signal strong enough that it would likely already be known.

- **The bottleneck is response-annotated cohorts, not methylation.** Methylation arrays are abundant in public archives; methylation *plus* RECIST response *plus* uniform TKI treatment is what is scarce. If the group has access to such a cohort, this analysis is ready to run against it unchanged — the script, gates and controls are fixed and hashed.

5 Reproducing this report

- Analysis: `paper-draft/02-revised-protocol-open-datasets/m7_tki_methylation.py`
- Config hash (written before any outcome was read): `623dff57160f7688...9562fb0c`
- Evidence: `evidence/` — config, gates from both runs, results, per-patient out-of-fold predictions, the 100 permutation AUROCs, audit addendum, run log, and `MANIFEST.sha256`
- Figure: `python3 make_figure.py`
- Source data: GEO GSE147377, series matrix and `GSE147377_AverageBeta_Matrix.csv.gz`, staged under `labels/lung/GSE147377/`